



Ibad Ur Rahman

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🌐 Ibad Ur Rahman Memon 🔄 Ibad-Ur-Rahman-Memon

Profile

Computer Systems Engineering graduate from Sukkur IBA University with hands-on experience in Artificial Intelligence, Machine Learning, Robotics, Embedded Systems, Full-Stack Development, and Cybersecurity fundamentals. Built 15+ academic and industry-oriented projects spanning Generative AI, Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), Computer Vision, Robotics (ROS2), Mobile Applications, ASP.NET Core, Flutter, Digital System Design, Operating Systems, and Embedded Systems. Completed a Generative AI internship focused on designing intelligent chatbot solutions using OpenAI, LangChain, Hugging Face, Google Gemini, FAISS, and Streamlit. Co-authored a peer-reviewed research publication in medical image analysis, demonstrating strong research aptitude and practical application of deep learning.

Recognized for combining hardware and software engineering to solve real-world problems through interdisciplinary projects, leadership roles, technical event organization, and academic excellence. Passionate about AI, Intelligent Systems, Software Engineering, Robotics, and Cybersecurity, with a strong interest in pursuing research-driven graduate studies and contributing to innovative engineering teams worldwide.

Technical Skills

Programming

Python • C • C++ • Java • C# • SQL • Assembly (x86) • Verilog HDL

Artificial Intelligence & Machine Learning

Generative AI • Machine Learning • Natural Language Processing (NLP) • Large Language Models (LLMs) • Retrieval-Augmented Generation (RAG) • Prompt Engineering • AI Chatbot Development • Computer Vision • Deep Learning Fundamentals • Information Retrieval • Semantic Search • Text Embeddings

AI Frameworks & Libraries

LangChain • OpenAI API • Google Gemini API • Hugging Face Transformers • FAISS • Sentence Transformers • Scikit-learn • NumPy • Pandas • Matplotlib • Streamlit

Full-Stack Development

ASP.NET Core (.NET 8) • REST APIs • MVC Architecture • Flutter • Firebase • Streamlit • HTML • CSS • JavaScript (Basic) • Responsive UI Development

Databases

MySQL • SQLite • Firebase Firestore • SQL Database Design

Robotics & Automation

ROS2 • LiDAR Navigation • Autonomous Mobile Robots (AMR) • Obstacle Avoidance • Sensor Integration • Robot Navigation • Mobile Robotics

Embedded Systems

Arduino • ESP32 • Raspberry Pi • Embedded Systems • Microcontrollers • Digital System Design • Finite State Machines (FSM) • Hardware-Software Integration

Networking & Cybersecurity

Linux • Windows • Operating System Concepts • Process Scheduling • Memory Management • Custom Operating System Development

Computer Vision & Image Processing

Image Restoration • Image Deblurring • Wiener Filter • MATLAB Image Processing • Medical Image Analysis

Software Engineering

Object-Oriented Programming (OOP) • Data Structures & Algorithms • Software Development Life Cycle (SDLC) • Version Control • Software Documentation • Debugging • Problem Solving

Tools & Technologies

Git • GitHub • Visual Studio • Visual Studio Code • Android Studio • MATLAB • Simulink • AutoCAD • Cisco Packet Tracer • Postman

Cloud & AI Platforms

OpenAI Platform • Google AI Studio • Google Gemini • Hugging Face Hub • Groq API

Research & Academic Skills

Research Methodology • Technical Writing • Scientific Literature Review • Academic Paper Writing • Experimental Evaluation • Model Performance Analysis • Citation Management

Soft Skills

Leadership • Team Collaboration • Project Management • Technical Communication • Public Speaking • Critical Thinking • Analytical Thinking • Problem Solving • Event Coordination • Time Management • Adaptability • Fast Learning • Mentoring

Professional Experience

06/2025 – 07/2025 Lahore, Pakistan (On-site)	TechInn Global  Generative AI Intern – Custom Chatbot Development Worked on AI-powered chatbot and NLP applications integrating OpenAI, Hugging Face, and Google Gemini APIs. Developed intelligent systems using LangChain, Streamlit, and FAISS for real-time information retrieval and automation. Key projects include ScholarBot , StyleHub AI , and Halal Restaurant Name Generator .
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Education

09/2022 – 06/2026 Sukkur, Pakistan	BE-Computer Systems Engineering Sukkur IBA University 
09/2020 – 05/2022 Islamabad, Pakistan	Pre-Engineering Dr. A. Q. Khan school System & Collage Ibrahim Campus 
01/2019 – 05/2020 Shikarpur, Pakistan	Matriculation Hira Public School 

Publications

12/03/2026

Gastrointestinal Tract Lesion Classification Using Deep Involution Neural Networks [↗](#)

Pakistan Journal of Medical & Cardiological Review

Co-authored a paper proposing a lightweight four-layer Deep Involution Neural Network for automated classification of GI tract lesions from endoscopic images into Normal, Ulcerative Colitis, Polyps, and Esophagitis. The model uses involution operations to extract features efficiently at low computational cost, achieving 91.13% accuracy on a 6,000-image public dataset — offering a lightweight, deployable alternative to complex conventional deep learning methods.

Projects

OmniMind – AI-Powered Mental Health Monitoring System [↗](#)

Final Year Project (FYP) | LLMs, RAG, NLP, Flutter, Firebase

AI-powered mental health monitoring system (Final Year Project) combining LLMs, RAG, and NLP with clinically validated assessments (PHQ-9, GAD-7, PSS-10, SPIN) to deliver personalized, evidence-based mental health support via a Flutter mobile app with real-time emotion detection and crisis intervention.

ResumeForge AI – AI-Powered Career Toolkit [↗](#)

Full-Stack Web App | ASP.NET Core, Groq LLaMA 3.1

Full-stack AI career toolkit built with ASP.NET [↗](#) Core 8 and Groq LLaMA 3.1, featuring a real-time resume builder, AI-driven ATS compatibility checker, and automated cover letter generation.

FYP-Assist – Project Management Platform [↗](#)

Developed a mobile application using Flutter and Kotlin to streamline the management of Final Year Projects (FYP) for university students and supervisors. Features automated project proposal tracking, task management, and repository linking to enhance academic collaboration. Built a robust backend to handle real-time project documentation and milestone tracking for university departments.

LiDAR-Reactive-Navigation – ROS2 Robotics [↗](#)

Developed a reactive navigation system for autonomous robots using ROS2 and LiDAR sensor integration. Implemented obstacle avoidance algorithms based on real-time sensor data, enabling the robot to navigate complex environments dynamically. Focused on mobile robot kinematics and sensor fusion to optimize autonomous path-planning and navigation efficiency.

Automatic Railway Gate Controller – DSD Project [↗](#)

Designed and simulated an automated railway safety system using Verilog HDL as part of a Digital System Design course. Implemented a Finite State Machine (FSM) to control gate operations based on sensor inputs, ensuring real-time responsiveness and mechanical reliability. Verified the hardware logic through extensive testbench simulations to handle diverse railway traffic scenarios and ensure safety standards.

ScholarBot – AI-Powered Research Assistant [↗](#)

Developed an AI-powered academic assistant that allows users to upload research PDFs or insert up to 3 article URLs, ask academic questions, and receive citation-aware responses in real time. Integrated OpenAI GPT-3.5, LangChain, FAISS, and sentence-transformers for intelligent document retrieval and summarization. Deployed using Streamlit for both local and Colab environments, enabling seamless academic exploration.

Halal Restaurant Name Generator – AI-Based Application [↗](#)

Developed an AI-powered web app that generates culturally appropriate halal restaurant names and traditional menu items based on Islamic countries. Built with Python, Streamlit, LangChain, and OpenAI API, featuring prompt-based regional authenticity and secure API handling through `.env` configuration. Showcases fusion of cultural heritage with modern AI technology.

StyleHub AI – Smart Inventory Q&A System [↗](#)

Created an AI-driven inventory management system that answers natural language queries about T-shirt stock using Google Gemini for SQL generation, MySQL for real-time queries, and Streamlit for web interface. Implemented NLP-based query handling, discount-aware revenue calculations, and cached responses for efficiency. Designed responsive UI and optimized query performance using Python's `lru_cache` and LangChain few-shot learning.

RealixOS – Mini Operating System

Developed a real-time OS featuring process management, memory management, and a CLI. Implemented a round-robin scheduler for efficient task execution. Presented in a departmental exhibition, aiming for top position.

Autonomous Mobile Robot for Corrosion Detection

Designed an AMR for underground pipeline inspection using Arduino. Implemented corrosion detection via ultrasonic sensors and a copper coil mechanism. Won **1st Position** at the **Tecnofun event at Shaikh Ayaz University**.

Network Security Implementation

(Cisco Packet Tracer)
Configured firewalls, VLANs, and access control lists for secure communication. Designed a network topology simulating a secure corporate network.

Image Deblurring and Restoration Using Wiener Filter

Simulated motion blur and deblurring processes, exploring the impact of Gaussian noise and noise estimation on image quality improvement using MATLAB.

3D Model of Malaysian Petronas Twin Towers

(AutoCAD)
Created a 3D model of the Petronas Towers as part of the CAD course, improving 3D modeling and architectural visualization skills.

CSE Search Engine – GUI-Based

Developed a search engine in Java as part of Object-Oriented Programming. Featured a **Google-inspired** GUI that displayed department student data upon CMS ID or name search.

Currency Converter– CLI-Based

Developed a real-time Currency Converter using C++ that supported conversions between multiple currencies (Dollar-Riyal, PKR-Dollar, etc.).

LED Cube – The Basis of Hologram Technology

Designed a 3D LED cube demonstrating hologram technology concepts and screen structuring.

Hydraulic Crane

(Physics Project)
Built a hydraulic crane using recycled materials, won **1st position** at a district-level project expo.

Certificates		
• Cisco Networking Essentials (Self-Learned) • Introduction to Ethical Hacking (By Great Learning) • System Administration and IT Infrastructure Services – Google • IT Security: Defense against the Digital Dark Arts – Google	• What is Generative AI (By LinkedIn Learning) • Foundations of Cybersecurity – Google • Crash Course on Python – Google • Introduction to AI – Google	• Generative AI: The Evolution of Thoughtful Online Search • Operating Systems and You: Becoming a Power User – Google • The Bits and Bytes of Computer Networking – Google • Foundations of Project Management – Google

Areas of Interest		
• Artificial Intelligence • Natural Language Processing • Software Engineering • Computer Vision	• Machine Learning • Robotics • Cybersecurity • Human-Computer Interaction	• Generative AI • Embedded Systems • Intelligent Systems • Research & Innovation

Experience & Extracurricular Activities

Freelancer on Fiverr

- Online Tutor (Math & Algebra), Graphic Designer, and Programmer.

Scholarships

Sindh Educational Endowment Fund Scholar

- Awarded full tuition coverage at Sukkur IBA University from 4th semester onward.

World Memon Organization Scholar

- Awarded full tuition coverage at Sukkur IBA University (1st-4th semester).

United Memon Jamat of Pakistan Scholar

- Awarded full tuition coverage at Dr. A.Q. Khan School System & College Islamabad during Intermediate.

Leadership Roles

Council Member – IBA University Students' Council (2023-24)

- Served as Coordinator of the Computer Systems Engineering Society, organizing SIBA Fest'24 and other technical events.

Team Lead – Pakistan Literature Festival 2023

- Led the Disciplinary team at a two-day literature event at Sukkur IBA University.

Student Volunteer – Career Fair 2024

- Assisted in organizing Career Fair 2024 as part of the Students' Council Disciplinary Team.

Event Coordinator – Sukkur IBA University Students' Council

- Organized farewell parties, project exhibitions, and technical seminars at Sukkur IBA University.

Student Volunteer, International Conference (March 2024)

- Assisted in the 5th International Conference on Business, Economics & Management at Sukkur IBA University.

Participant – Project Exhibition (CSE Department, SIBA)

- Presented a project under the Engineering Workshop domain.

1st Position – Hira Public School

(Since 8th Class)

- Consistently ranked 1st in academics since 8th grade.

Awards

Best Organizer Award (SIBA Fest'24)

Sukkur IBA University

1st Position – Project Competition (Tecnofun)

The Shaikh Ayaz University Shikarpur

1st Position – Hydraulic Crane (Project)

Shikarpur Pre-Stage Club

Ja'an Nisar Award – Volunteering at PLF Chapter-1

Art Council of Pakistan Karachi

Best Performance Award

Hira Public School (Bazm-e-Sathi, Jamat-e-Islami)

Tech-Quiz Participant

Galaxy School, Shikarpur

Languages

English



Urdu



Sindhi



Arabic



References

Muhammad Mudassar Qureshi, CEO, Tech Inn Global

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